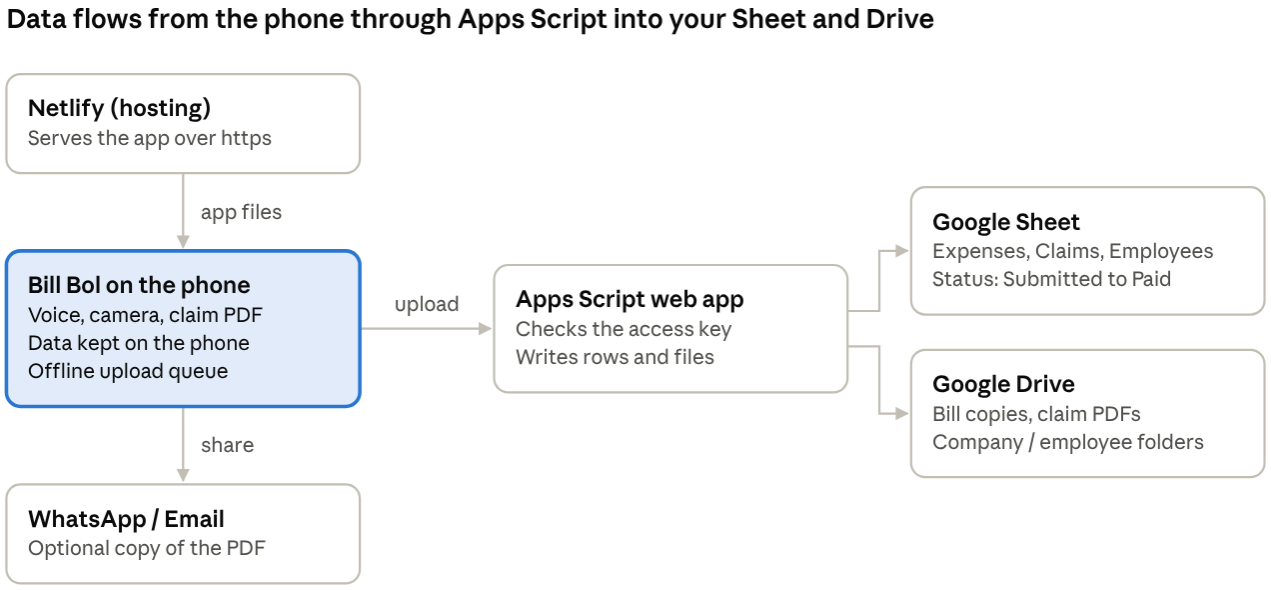
Bill Bol — Programs and Technologies Used

Oct 22, 2026 · @Vishal

Bill Bol is written in HTML, CSS and JavaScript (the app), Google Apps Script (the database) and Python (build and helper tools). It uses only free, open-source libraries and the phone's own voice, camera and sharing features. The complete source code is submitted with this document as **Bill-Bol-Source-Code.zip**.

Architecture



Bill Bol architecture · phone, Apps Script, Sheet, Drive

The phone does all the work (voice, photos, PDF) and keeps the data; the Apps Script web app only receives uploads and writes them into the firm's own Sheet and Drive. No separate server or paid database is needed.

Program files written for the application

File	Language	Size	Purpose
app/bill-bol.html	HTML, CSS, JavaScript	1,491 lines	The complete application: screens, voice understanding, bill capture, claim PDF and Excel, sharing, sync with the company database
google-sheet-database/BillBol-GoogleSheet-Database.gs	Google Apps Script (JavaScript)	246 lines	Web app that writes entries, claims and employees into Google Sheets and saves bill and claim files in Google Drive
build/build-bill-bol-site.py	Python 3	100 lines	Builds the deployable website: bundles libraries, creates app icons, manifest and offline service worker, zips the folder
website/sw.js	JavaScript (service worker)	15 lines	Offline use and automatic updates of the installed app
website/manifest.webmanifest	JSON	37 lines	App name, icons and colours, so Android and iPhone can install it
website/index.html	HTML, CSS, JavaScript	1.9 MB	The deployable app with PDF, PDF-merge and Excel libraries bundled for offline use
optional-laptop-mail-helper/bol-khata-helper.py	Python 3	364 lines	Optional: sends email with attachments from a computer through a Gmail app password
optional-laptop-mail-helper/start-bol-khata-helper.bat	Windows batch	9 lines	Starts the mail helper on Windows

The main program contains 97 functions in two parts: an **engine** of pure calculation functions (speech parsing, totals, journal entry, amount in words) and the **app** layer (screens, storage, PDF, sync).

Languages, libraries and platform services

Component	Version	Used for	Licence / cost
HTML5, CSS3, JavaScript (ES2020)	Browser standard	All screens and logic	Open standard
jsPDF	2.5.1	Creating the claim PDF	MIT, free
jsPDF-AutoTable	3.8.2	Tables in the claim PDF	MIT, free
pdf-lib	1.17.1	Merging PDF invoices into the claim	MIT, free
SheetJS (xlsx)	0.18.5	Excel export	Apache 2.0, free
Inter font	Google Fonts	Interface typography	SIL Open Font Licence, free
Web Speech API	Browser built-in	Voice input and spoken questions (English, Hindi, Tamil)	Free, built into Chrome and Safari
IndexedDB and local storage	Browser built-in	Saving entries, bill photos and signature on the phone	Free
Camera access and file input	Browser built-in	Taking bill photos on phone and laptop	Free
Web Share API	Browser built-in	Sending the PDF to WhatsApp or email	Free
Service worker and web app manifest	Browser built-in	Install as app, offline use	Free
Google Apps Script	Google Workspace	Database web app	Free with a Google account
Google Sheets and Google Drive	Google Workspace	Expense, claim and employee records; bill and claim files	Free within Google storage limits
Netlify	Free plan	Hosting the website over https	Free plan

Component	Version	Used for	Licence / cost
Python 3 with Pillow	3.x	Build script and app icons	PSF / HPND licences, free

Key modules inside the main program

Module	Main functions	What it does
Speech understanding	parseExpense, tokenize, findNumbers	Reads amount (digits and number words in English, Hindi, Tamil), category, date, payment mode, place and route from one spoken sentence; lists what is missing
Question engine	missing, askNext, speak	Asks only for missing details, by voice and on screen; never guesses
Voice capture	startMic, micAvailable	Uses the phone's speech recognition; falls back to the keyboard mic on iPhone
Bill capture	openCamera, compress, readPdfBill	Camera photos, gallery images and PDF invoices; compresses and stores them on the phone
Claim calculations	claimSummary, journalEntry, amountInWords, claimNo	Totals, advance, net payable, account heads, suggested journal entry, claim number
Claim PDF	drawSheet, buildPDF, mergeBills	One-page claim form with font auto-fit, bill pages, merged PDF invoices, page numbers
Excel export	buildXLSX	Claim, summary and journal sheets
Sharing	sendClaim, sendModal	WhatsApp or email through the phone's share sheet, or download with a prefilled message

Module	Main functions	What it does
Company database sync	enqueue, syncNow, cloudPost, submitClaim	Offline queue; uploads entries, bill files and claims to the Google Sheet web app with retry
Profile and setup	viewOnboarding, saveProfile, applySetupLink	Profile-first lock; setup link from the accountant connects the phone to the firm's sheet
Install and checks	installCard, runDiag	Install prompt for Android and iPhone; phone check report

Database web app (Apps Script): doPost receives the data and checks the access key; saveEntry_ writes the Expenses row and Drive files; saveClaim_ writes the Claims row, the PDF and the Claimed status; upsertEmployee_ maintains the Employees sheet; setup creates the sheets, the status dropdown, the Drive folder and the access key.

Development and testing tools

Tool	Used for
Claude (Anthropic AI assistant)	Writing all code from plain-language prompts; testing and packaging
Node.js with jsdom	Automated tests of the speech parser, claim maths, PDF and Excel generation
Playwright with Chromium	Automated phone-screen tests (Android and iPhone sizes), install checks, offline checks, screenshots
Local Apps Script stand-in (Node.js)	Running the real Apps Script code against simulated Sheets and Drive to test the database end to end
Poppler (pdftoppm, pdfinfo)	Checking claim PDF pages and page counts
Microsoft Edge, Chrome, Safari	Manual testing on laptop, Android phone and iPhone by the developer